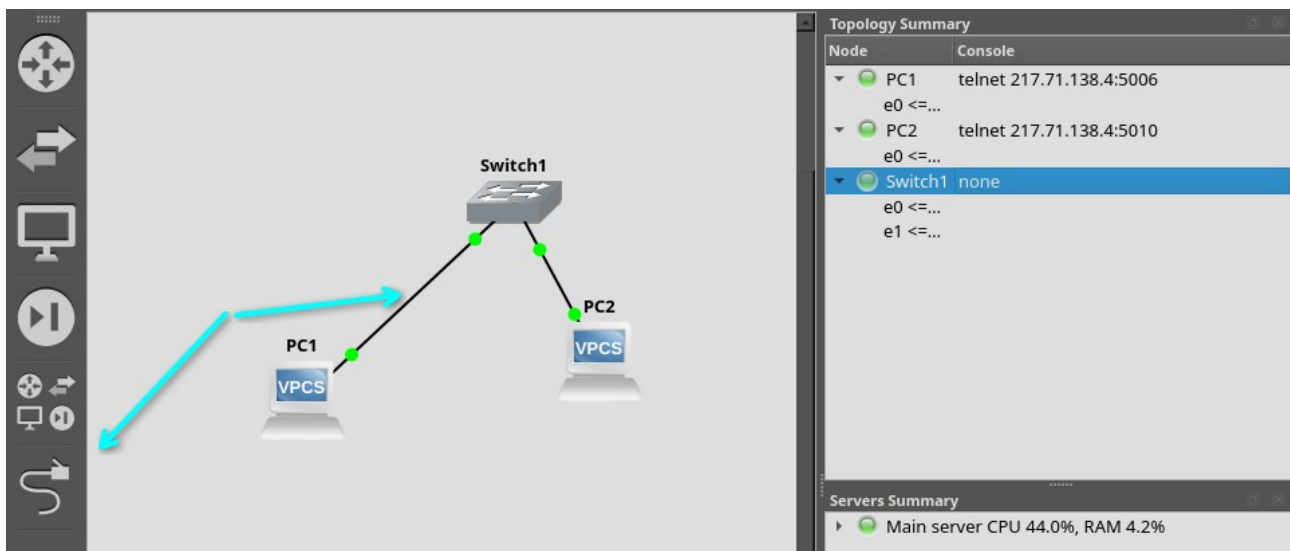


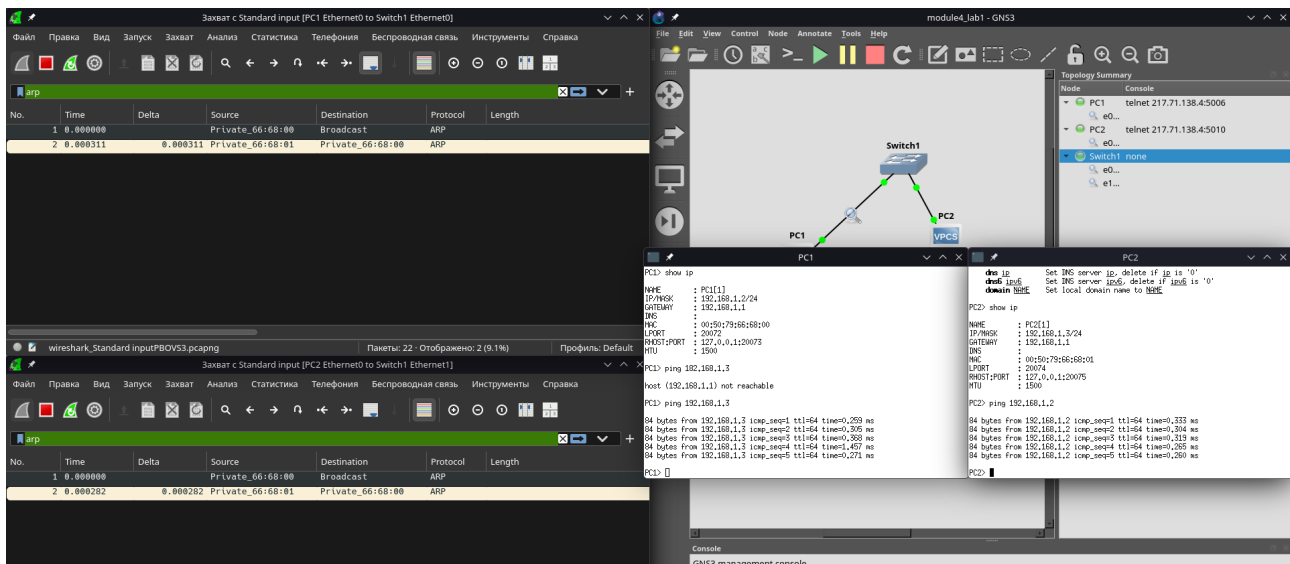
Module 4 lab 1

Корначенко Данил Александрович

Адрес задаётся :> ip <адрес> <маска> <gateway/шлюз>

PC1	PC2
Trying 217.71.138.4... Connected to 217.71.138.4. Escape character is '^['. Checking for duplicate address... PC1 : 192.168.1.2 255.255.255.0 PC1> ip 192.168.1.2/24 192.168.1.1 Checking for duplicate address... PC1 : 192.168.1.2 255.255.255.0 gateway 192.168.1.1 PC1> show ip NAME : PC1[1] IP/MASK : 192.168.1.2/24 GATEWAY : 192.168.1.1 DNS : MAC : 00:50:79:66:68:00 LPORT : 20072 RHOST:PORT : 127.0.0.1:20073 MTU : 1500 PC1> █	of the subnet. In the example above the tapx ip would be 10.1.1.126 mask may be written as /26, 26 or 255.255.255.192 Attempt to obtain IPv6 address, mask and gateway using SLAAC Attempt to obtain IPv4 address, mask, gateway, DNS via DHCP -d Show DHCP packet decode -r Renew DHCP lease -x Release DHCP lease dns ip Set DNS server ip, delete if ip is '0' dns6 ipv6 Set DNS server ipv6, delete if ipv6 is '0' domain NAME Set local domain name to NAME PC2> show ip NAME : PC2[1] IP/MASK : 192.168.1.3/24 GATEWAY : 192.168.1.1 DNS : MAC : 00:50:79:66:68:01 LPORT : 20074 RHOST:PORT : 127.0.0.1:20075 MTU : 1500 PC2> █





Source — откуда идёт пакет, Destination — куда идёт
 ARP нужен чтобы узнать MAC адрес устройства по ip и идёт первым.

Ставлю маршрутизатор и настраиваю.

```
R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#show interface fastEthernet0/0
^
% Invalid input detected at '^' marker.

R1(config)#interface fastEthernet0/0
R1(config-if)#ip address 192.168.12.2 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:05:51.791: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:05:52.791: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#interface fastEthernet1/0
R1(config-if)#ip address 192.170.12.2 255.255.255.0
^
% Invalid input detected at '^' marker.

R1(config-if)#ip address 192.170.12.2 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar 1 00:07:05.867: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 00:07:06.867: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config)#end
R1#
*Mar 1 00:07:10.127: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

PC1

```
PC1> ip 192.168.12.1/24 192.168.12.2
Checking for duplicate address...
PC1 : 192.168.12.1 255.255.255.0 gateway 192.168.12.2

PC1> show ip

NAME           : PC1[1]
IP/MASK         : 192.168.12.1/24
GATEWAY         : 192.168.12.2
DNS             :
MAC             : 00:50:79:66:68:00
LPORT          : 20960
RHOST:PORT      : 127.0.0.1:20961
MTU             : 1500

PC1> 
```

PC2

```
PC2> ip 192.170.12.1/24 192.170.12.2
Checking for duplicate address...
PC2 : 192.170.12.1 255.255.255.0 gateway 192.170.12.2

PC2> show ip

NAME           : PC2[1]
IP/MASK         : 192.170.12.1/24
GATEWAY         : 192.170.12.2
DNS             :
MAC             : 00:50:79:66:68:01
LPORT          : 20962
RHOST:PORT      : 127.0.0.1:20963
MTU             : 1500

PC2> 
```

Пингуем

```
PC1> ping 192.170.12.1

192.170.12.1 icmp_seq=1 timeout
84 bytes from 192.170.12.1 icmp_seq=2 ttl=63 time=11.473 ms
84 bytes from 192.170.12.1 icmp_seq=3 ttl=63 time=16.600 ms
84 bytes from 192.170.12.1 icmp_seq=4 ttl=63 time=16.098 ms
84 bytes from 192.170.12.1 icmp_seq=5 ttl=63 time=16.053 ms

PC1> 
```

Захватываем с помощью wireshark

Захват с Standard input [PC1 Ethernet0 to R1 FastEthernet0/0]							
No.	Time	Delta	Source	Destination	Protocol	Length	
4	21.984937		Private_66:68:00	Broadcast	ARP		
5	21.919315	0.014378	cc:01:e0:21:00:00	Private_66:68:00	ARP		
6	21.919890	0.000575	192.168.12.1	192.170.12.1	ICMP		
7	23.920630	2.000740	192.168.12.1	192.170.12.1	ICMP		
8	23.932004	0.011374	192.170.12.1	192.168.12.1	ICMP		
9	24.932398	1.000394	192.168.12.1	192.170.12.1	ICMP		
10	24.948912	0.016514	192.170.12.1	192.168.12.1	ICMP		
11	25.949455	1.000543	192.168.12.1	192.170.12.1	ICMP		
12	25.965439	0.015984	192.170.12.1	192.168.12.1	ICMP		
13	26.966514	1.001075	192.168.12.1	192.170.12.1	ICMP		
14	26.982456	0.015942	192.170.12.1	192.168.12.1	ICMP		

wireshark_Standard input5NACT3.pcapng Пакеты: 24 - Отображено: 11 (45.8%) Профиль: Default

Захват с Standard input [PC2 Ethernet0 to R1 FastEthernet1/0]							
No.	Time	Delta	Source	Destination	Protocol	Length	
3	11.380555		cc:01:e0:21:00:10	Broadcast	ARP		
4	11.380555	0.000100	Private_66:68:01	cc:01:e0:21:00:10	ARP		
5	13.373084	1.992429	192.168.12.1	192.170.12.1	ICMP		
6	13.373204	0.000120	192.170.12.1	192.168.12.1	ICMP		
7	14.390022	1.016818	192.168.12.1	192.170.12.1	ICMP		
8	14.390129	0.000107	192.170.12.1	192.168.12.1	ICMP		
9	15.406527	1.016398	192.168.12.1	192.170.12.1	ICMP		
10	15.406669	0.000142	192.170.12.1	192.168.12.1	ICMP		
11	16.423562	1.016893	192.168.12.1	192.170.12.1	ICMP		
12	16.423725	0.000163	192.170.12.1	192.168.12.1	ICMP		

wireshark_Standard inputJ5U653.pcapng Пакеты: 21 - Отображено: 10 (47.6%) Профиль: Default